DUST on the K=32 code-carried model: the 500-step $\sigma\times$ `dust_lr_scale` grid

Research note

2026-10-22.

Single runs, seed 1337; the encoder arm and the baseline trained on new_york (8× RTX 5090), the backprop arm on georgia (8× A100).

1 Setup

The model is the K=32 code-carried hard Top-32 model of `docs/kj-vs-hard-topk-code-residual.tex` (`configs/cr_kj/ma_cr_hard_k32_pnorm.yaml`): 8 blocks of width 768, abs-TopK over $N = 1536$ features at `residual_out` of every block, the code carried between blocks, post-norm, magnitude–direction (MD) decoupling, batch 96×512 tokens, a 20k-step cosine schedule with 500 warmup steps.

Every DUST run (`configs/cr_dust/hp500/`, method in `docs/dust-topk.md`) adds the DUST bottleneck optimization: the $J = 480$ pool receives the noise, population 128 (16 draws per bottleneck), schedule $[1, 1]$ (every other optimizer step a DUST step), $\gamma = 0$, gradient scale 1, a separate optimizer for the DUST steps sharing the MD parameterization, and no gradient clipping. The grid is `dust_target` $\in \{\text{encoder}, \text{backprop}\} \times \sigma \in \{0.03, 0.1, 0.3, 0.9\} \times$ `dust_lr_scale` $\in \{1, 10, 30, 100, 300\}$. Each run trains 500 steps of the unchanged 20k-step schedule, so the window lies inside the LR warmup ($1.2 \cdot 10^{-6}$ to $6 \cdot 10^{-4}$; the DUST steps use that value times `dust_lr_scale`), with validation every 50 steps on 40 batches of 24 windows.

The baseline is the same model trained with the same settings and no DUST steps (`ny_hp500_hard_k32`); the original madrid run of the model (gradient clipping at 1.0) is quoted for reference.

2 Encoder target

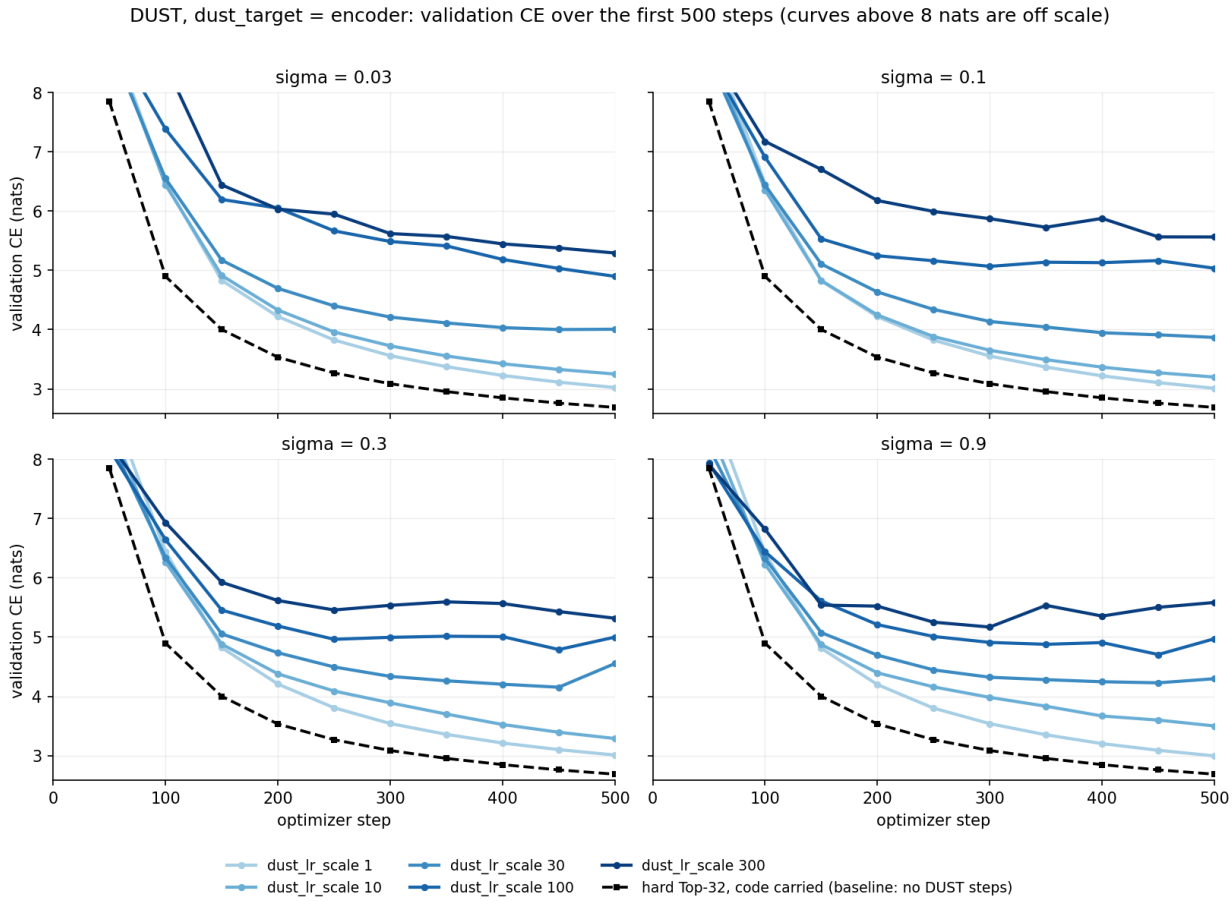


Figure 1: `dust_target = encoder`: validation CE over the first 500 steps, one panel per σ, one curve per `dust_lr_scale` (one hue, light to dark with the multiplier); dashed black, the hard Top-32 baseline. Curves above 8 nats are off scale.

Validation CE at step 500 (bold: the best cell):

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.0220	3.2500	4.0055	4.8971	5.2906
0.1	3.0096	3.1989	3.8673	5.0334	5.5611
0.3	3.0101	3.2891	4.5535	4.9989	5.3168
0.9	**2.9957**	3.5009	4.2972	4.9715	5.5825
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

3 Backprop target

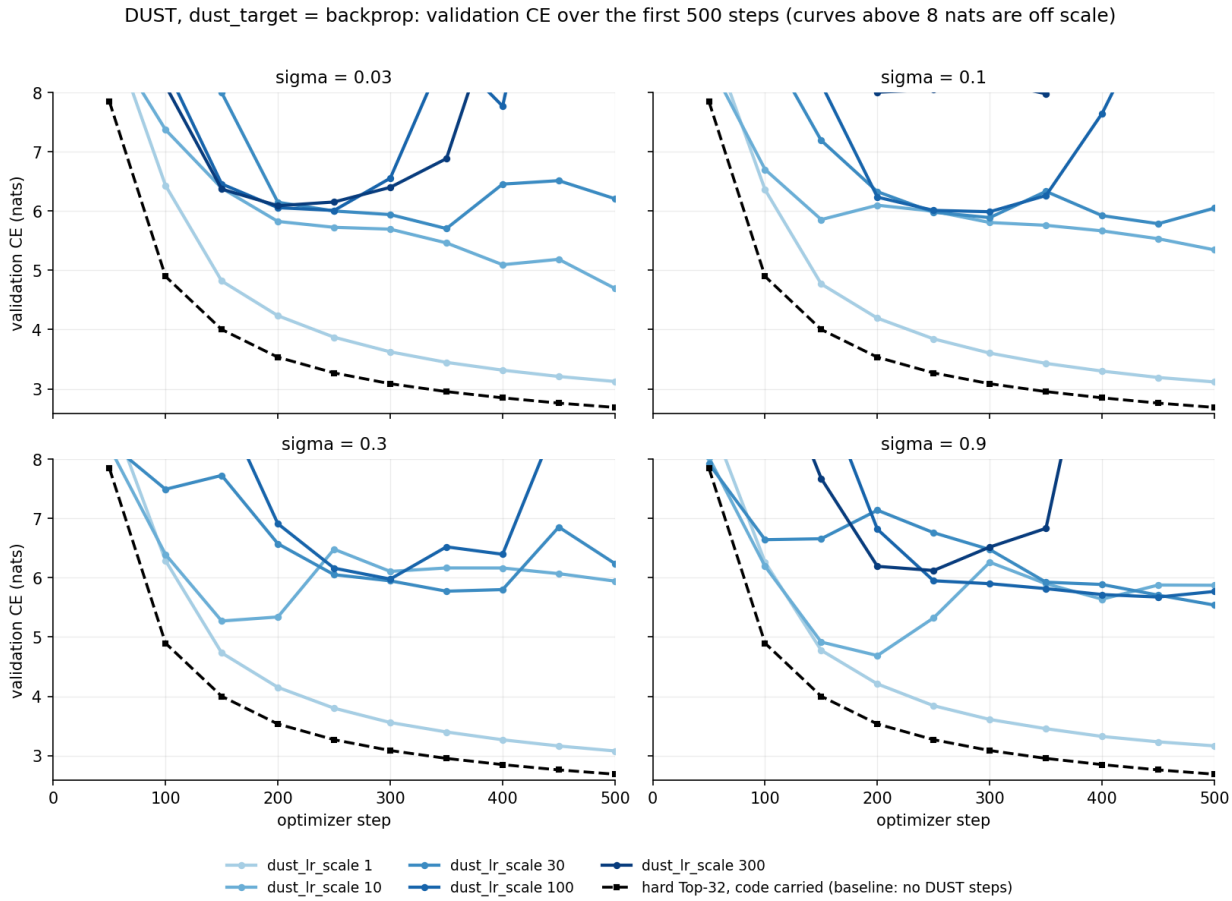


Figure 2: `dust_target = backprop`: the same layout. Curves above 8 nats are off scale.

Validation CE at step 500:

σ \ `dust_lr_scale`	1	10	30	100	300
0.03	3.1246	4.6930	6.2055	13.0487	9.7345
0.1	3.1169	5.3437	6.0478	8.8164	9.9774
0.3	**3.0794**	5.9428	6.2382	11.9248	9.6194
0.9	3.1646	5.8732	5.5386	5.7687	9.2661
hard Top-32 baseline (no DUST steps, same settings)	2.6875				
hard Top-32, the original madrid run (clipping 1.0)	2.6846				

4 Results

No cell reaches the baseline. In both arms the validation CE at step 500 rises monotonically with `dust_lr_scale`, and the multiplier 1 is the best value at every σ: 2.996 to 3.022 in the encoder arm, 0.31 to 0.33 nats above the baseline, and 3.08 to 3.16 in the backprop arm. The dependence on σ is weak at the multiplier 1 (a spread of 0.03 nats in the encoder arm and 0.09 in the backprop arm) and does not change the ordering by the multiplier anywhere.

The two targets differ in how the larger multipliers fail. In the encoder arm the curves keep descending at every setting, only more slowly and from a higher level as the multiplier grows; at 100 and 300 they plateau near 5 to 5.6 nats within the window. In the backprop arm, where the

estimate is backpropagated to every parameter before the gate inputs, every multiplier from 10 on stalls near 5 to 6 nats, and 100 and 300 diverge at every σ (final values of 5.8 to 13 nats), with the pre-activation scale growing by orders of magnitude within the first tens of steps.

Within the first 500 steps the DUST steps therefore cost against hard steps at every setting of the grid, least when the DUST learning rate is the schedule's own, and the encoder target is less harmful than the backprop target at every σ and multiplier.

5 Sigma sweep at population 1024

Pending: the eight runs (`dust_lr_scale` 1, population 1024 = 128 draws per bottleneck, both targets, the four σ values, otherwise the grid's settings) are training on new_york; their two figures and table will be added here.

6 Without MD decoupling

Pending: the hard Top-32 run without MD decoupling (`model.decouple = false` with `model.md_init = true`, so the initialization is identical and only the optimizer changes: ordinary AdamW with weight decay 0.1, the convention of `docs/md-init-vs-decoupling.tex`), the population-1024 σ sweep without decoupling at $\gamma = 0$ (georgia) and at $\gamma = 0.99$ (new_york). To be added: the hard run with and without decoupling; and, per γ, a two-panel figure (encoder left, backprop right) of the sweep against the hard run without decoupling, with its table.

7 Runs on Drive

- Encoder arm, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_enc_sg<s>_lr<l>` for `<s>` $\in$ {`0p03`, `0p1`, `0p3`, `0p9`} and `<l>` $\in$ {1, 10, 30, 100, 300}, twenty runs.
- Baseline, new_york: `gdrive:weight-sparsity/runs_new_york/ny_hp500_hard_k32`.
- Backprop arm, georgia: `gdrive:weight-sparsity/runs_georgia/ga_hp500_bp_sg<s>_lr<l>`, the same twenty cells.
- Original hard Top-32 run (reference value only): `gdrive:weight-sparsity/runs_madrid/ma_cr_hard_k32_pnorm`.

Figures and tables: `scripts/plot_dust_hp500.py` over the `metrics.jsonl` of these runs. The Markdown version of this note, `docs/dust/hp500-sigma-lr.md`, lists every run name in full.